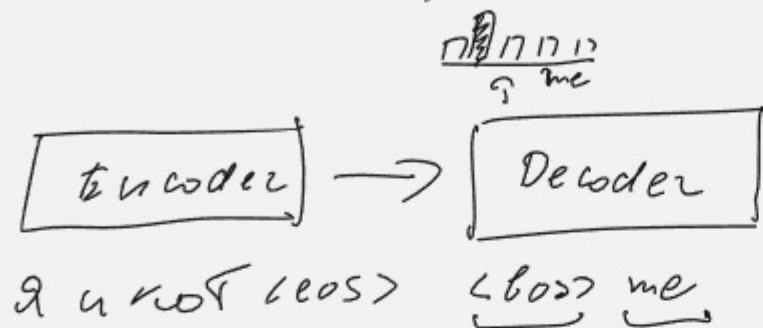


Teacher forcing

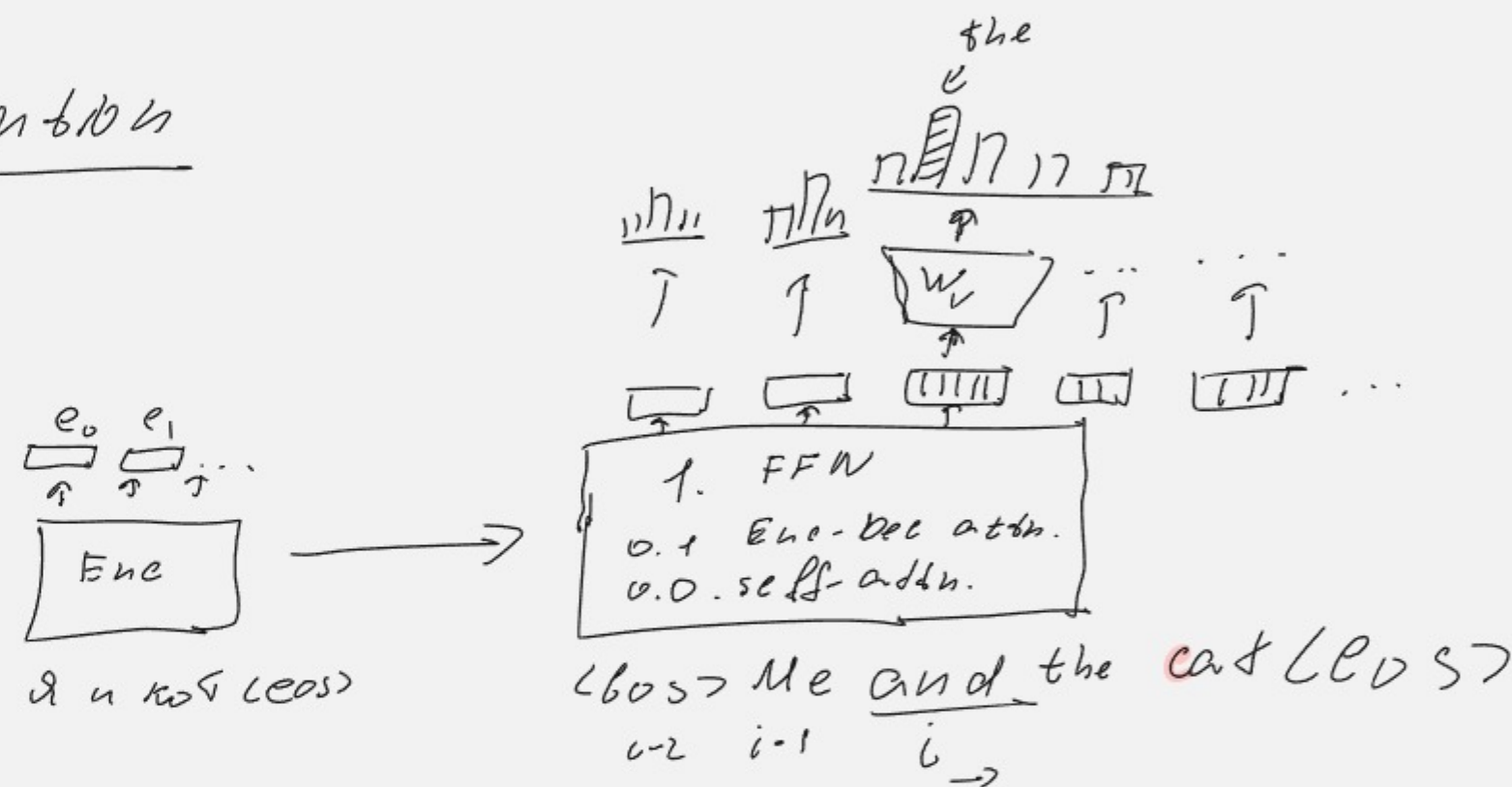
Example: Me (and) the cat.

CE $\rightarrow (000...1...0)$
 $(0.5d, 0.13... 0.23)$



Masked attention

input / output
 a u not / me and the cat



$$\begin{bmatrix} \langle \text{bos} \rangle \\ \text{me} \\ \text{and} \end{bmatrix} W_Q \rightarrow \begin{bmatrix} q(\langle \text{bos} \rangle) \\ q(\text{me}) \\ q(\text{and}) \end{bmatrix} = Q$$

$$\begin{bmatrix} e_0 \\ e_1 \\ e_2 \end{bmatrix} \rightarrow W_K \rightarrow \underline{K}$$

$$\begin{bmatrix} e_0 \\ e_1 \\ e_2 \end{bmatrix} \rightarrow W_V \rightarrow \underline{V}$$

Masked attention

$$Q \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}_{3 \times d} \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}_{e \cdot (\text{bos})} = \begin{bmatrix} 0 & 1 & 2 \\ \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} \end{bmatrix}_{3 \times 3}$$

$$\left(\text{softmax} \left(\frac{QK^T + \text{mask}}{\sqrt{d}} \right) V \right)$$

$$\text{mask} = \begin{bmatrix} 0 & 0 & -\text{inf} \\ 0 & 0 & 0 \end{bmatrix}$$